

SCS2315 - ELECTRONIC AND PHYSICAL COMPUTING

IC & Logic Gates Experiment – 8

February 11, 2026

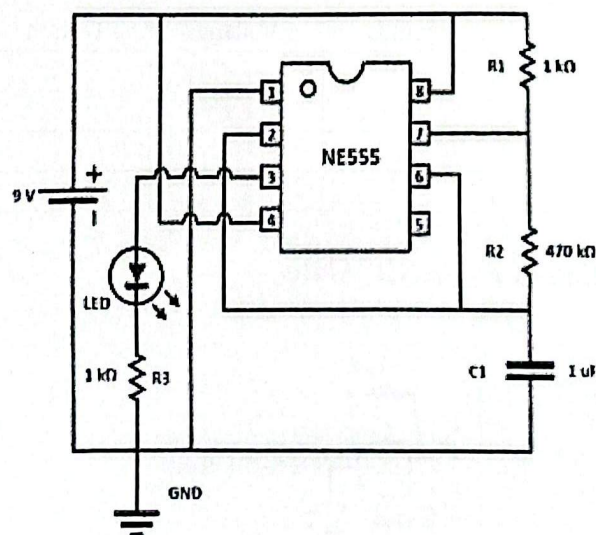
Objective

The blinking LED circuit using a 555 timer is designed to generate a continuous square-wave signal in a stable mode: the capacitor charges through two resistors, reaches $2/3 V_{CC}$, triggers discharge, falls to $1/3 V_{CC}$, and repeats making the LED blink at a rate set by R_1 , R_2 , and C . A NOT gate (inverter) flips its input outputting 0 for an input of 1 and vice versa—achievable via a single transistor or in integrated form (e.g., 74LS04). An AND gate outputs HIGH only when all inputs are HIGH, implementable using series transistors or simple diode logic (DTL) where diodes and resistors form the gate and a transistor provides inversion/amplification. NAND gates, the inverse of AND, are universal and can be used to build any other gate type.

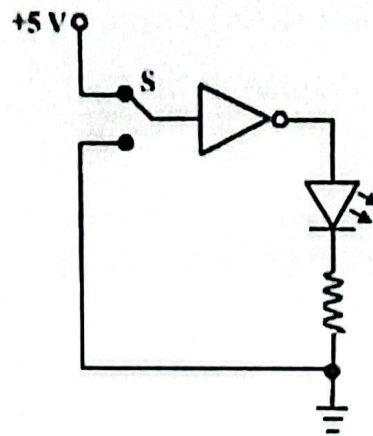
Materials

- NE555
- Resistors
- 9V Battery
- Capacitors
- LED
- Jumper wires
- Breadboard

Activity 1: Blinking LED Circuit Using 555 Timer IC



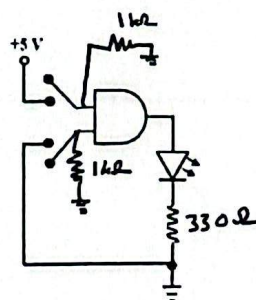
Activity 2: Operation of a “NOT” Gate



1. If you are using a 5 V power supply why would you need to connect a resistor in series with a LED before powering up.
2. How would you determine a suitable value of resistance to be used when testing the operation of a LED? Determine a suitable value of Resistance for the LED and the power supply you are given.
3. What component in Tinkercad can be used as component “S”
4. What 7400 series IC would you pick to construct the circuit in Figure
5. Where will you place the IC on the breadboard? Explain why?
6. What document would you need to determine the pin configuration of the IC you picked?
7. Construct the circuit shown in above Figure in Tinkercad.
8. Now confirm the function of the NOT gate. Report your observations in the following table. Measure output voltages and input voltages and include the voltage values also in the table.

Input A	Voltage	Input B	Voltage	Output	Voltage
0		0			
1		0			
0		1			
1		1			

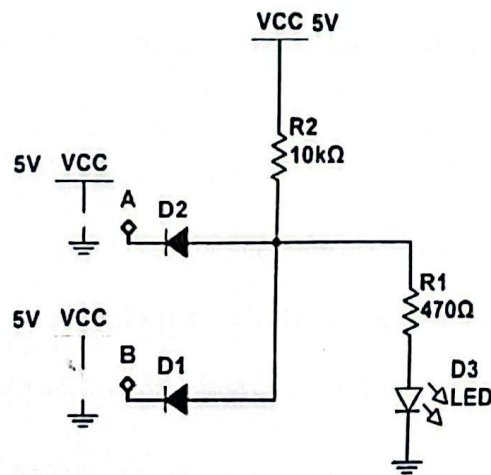
Activity 3: Operation of an “AND” Gate



1. What 7400 series IC would you pick to construct the circuit in Figure.
2. Construct the circuit shown in Figure.
3. Now confirm the function of the AND gate. Report your observations in a truth table. Measure output voltages and input voltages and include the voltage values also in the truth table.
4. Repeat the activities given in the exercise for the other types of gates OR, NAND, NOR, Exclusive-OR(EOR) and Exclusive NOR (EXNOR).

Activity 4: Development of Digital Gates Using Diodes and BJT's - Diode Logic and Gate

1. Construct the circuit as shown in the figure and Record the output voltages for the input levels.

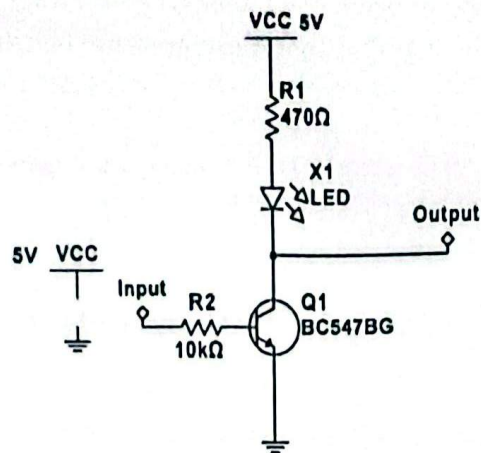


2. Report your observations in the following table. Measure output voltages and input voltages and include the voltage values also in the table.

Input A (V)	Input A (V)	Output (V)
0	0	
0	5	
5	0	
5	5	

3. Using the data in table construct the truth table

Activity 5: Transistor Logic NOT Gate



1. Construct the circuit as shown below and Record the output voltages for the input levels.

Input (V)	Output (V)
0	
5	

2. Using the data in the table, construct the truth table of the OR gate.

Activity 6: Ic-Based Gate Circuit Analysis - Not Gate Ic

1. Using 74LS04 IC, measure output voltages with respect to each input voltage level given in table.

Input (V)	Output (V)
0	
0.5	
1.0	
1.5	
2.0	
2.5	
3.0	
3.5	
4.0	
4.5	
5.0	

Activity 7: Ic-Based Gate Circuit Analysis - AND Gate Ic

Using 74LS08 IC, fill the table below by applying high and low input combinations to the two logic inputs.

A	B	Output (V)

Activity 8: Ic-Based Gate Circuit Analysis - NAND Gate Ic

Using 74LS00 IC, fill the table below by applying high and low input combinations to the two logic inputs

A	B	Output (V)